

SAI KARTHIK KOSURI

Philadelphia, PA · karthikjobasap@gmail.com · 6806974150 · <https://www.linkedin.com/in/karthikkosuri9/>

SKILLS

Programming:	Python, R, SQL, Go, Java, C++, JavaScript, Bash
Data Science:	Scikit-learn, TensorFlow, PyTorch, Hugging Face, MLflow, NLTK
Cloud/DevOps:	AWS, OpenAI, Claude, Databricks, Kubernetes, Docker, Jenkins, Terraform
Databases:	MySQL, MongoDB, Oracle, Teradata, Redis Cache
Analytics:	Power BI, Qlik Sense, Excel, statistical analysis (SciPy, hypothesis/A-B testing, model evaluation)

EXPERIENCE

University of Pennsylvania

Data Scientist

Philadelphia, PA

June 2024 - Present

- Developed a Retrieval-Augmented Generation (RAG) system by collecting 45K PDF articles from multiple sources, converting them to Markdown, chunking the text, generating embeddings with lightonai/GTE-ModernColBERT-v1, and storing them in a Weaviate vector database, integrated with an MCP server for query handling.
- Built multimodal ML framework combining linguistic (LIWC, LDA, n-grams) and acoustic (OpenSMILE, Librosa, Whisper) features to analyze 1,465 semi-structured interviews (310 participants, ages 60–124), achieving 34% correlation in predicting loneliness in older adults and identifying demographic-specific speech and language markers for early detection in senior care and telehealth applications.
- Developing agentic AI system for automated clinical living guidelines extraction, leveraging multi-agent workflows with large language models (Claude Opus 4, GPT-5) and NLP pipelines to autonomously process medical documents (RCTs) and extract structured clinical recommendations, reducing manual curation time by 75% and improving guideline accessibility for healthcare practitioners.
- Created a end-to-end data pipeline for participant longitudinal study using Python ETL (pandas, SQLAlchemy) to integrate Qualtrics surveys, secure file uploads, and social media data into normalized MySQL database. Automated survey scheduling and payment workflows, eliminating manual processes.
- Analyzed behavioral and psychometric data across 6 validated instruments for large participants, achieving 88%+ retention at 1-month follow-up. Performed statistical analyses (correlation, regression) linking social media activity to mental health outcomes and delivered actionable insights through data visualizations.

Siemens Healthineers

Data Science and Analytics Intern

Cary, NC

June 2023 - August 2023

- Engineered comprehensive business intelligence migration from Power BI to Qlik Sense, developing automated reporting workflows and advanced data visualization dashboards that improved analytics capabilities by 30% while implementing cost-optimization strategies that delivered \$50K in annual operational savings.
- Built automated ETL pipeline using Python, SQL, and Teradata to process enterprise healthcare datasets, reducing data processing errors by 50% and establishing scalable workflows for enhanced data quality and analytics applications.

Broadridge Financial Solutions

Software Engineer

Hyderabad, India

April 2021 - July 2022

- Developed robust REST APIs using Node.js with strategic optimization techniques, achieving 30% increase in user engagement through improved data delivery and processing efficiency for web applications.
- Automated infrastructure deployment via Jenkins CI/CD pipelines with Terraform IaC, reducing provisioning time by 70% and establishing scalable, reproducible environments for data processing and analytics workloads.
- Built scalable data processing pipeline using Go backend with concurrent routines to fetch, process, and store real-time API data streams in AWS S3, reducing data retrieval times by 30% and improving system responsiveness for analytics applications.
- Engineered automated monitoring and analytics infrastructure by implementing Splunk dashboards and CloudWatch logging systems, reducing root cause analysis time by 80% and enabling data-driven debugging and performance optimization.

PROJECTS

Deployment of Real-Time Voting Application on AWS

Utilized AWS EKS for backend orchestration and EC2 for React frontend hosting, ensuring enhanced performance, security & user experience with effective access policies and ALBs. Established MongoDB stateful set on EKS for data management

Covi source: Hospital Finder & Booking App

Designed and developed a Covi source (responsive desktop application for Tab and Mobile): encompassing full-stack design that enabled users to locate hospitals based on specific search criteria and make online reservations during the pandemic

EDUCATION

Syracuse University

Master's Computer Science *GPA: 3.8*

Syracuse, NY

Aug 2022 - May 2024

Vellore Institute of Technology

B.Tech Computer Science & Engineering *GPA: 8.49*

Vellore, India

July 2017 - June 2021

CERTIFICATIONS

AWS Certified Solutions Architect Associate